

OKELLO ELLY

Software Developer and Systems Architect specializing in scalable systems, full-stack engineering, and cybersecurity

CONTACT INFORMATION

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PROFILE

Software Developer and Systems Architect with hands-on experience designing and building secure, scalable software systems across digital learning, campus services, analytics, civic platforms, and operational workflows. Skilled in full-stack development, backend engineering, REST API design, database architecture, mobile application development, networking, cybersecurity, and system design, with practical implementation of JWT-based authentication and role-based access control (RBAC).

Experienced in applying software engineering principles, clean architecture, and systems thinking to develop reliable digital solutions. Strong foundation in software development, computer networking, data science, system configuration, software architecture, and cybersecurity, complemented by experience in AI-assisted development and data-driven problem solving.

Passionate about building technology that solves real-world challenges while leveraging emerging technologies, modern engineering practices, and scalable system design to create impactful software solutions.

EDUCATION

Bachelor of Science in Computer Software Engineering

Zetech University

Jan 2025 – Nov 2027

Core Technical Focus

- Software Development (Python, JavaScript, Java, C++, C)
- Full-Stack Web Development (HTML, CSS, JavaScript, REST APIs)
- Mobile Application Development (Java, Kotlin, Android Application Design)
- Visual Programming and Desktop Applications (C#, ASP.NET, Windows Forms)

- Database Design and Management (MySQL, SQLite, PL/SQL)
 - Backend Engineering and Secure API Development
 - Computer Networking (IP Addressing, VLANs, Routing, Switching, Network Configuration)
 - Cybersecurity and System Security Fundamentals
 - Data Science and Analytics with Python
 - System Analysis, Design, and Software Architecture
 - Software Metrics, Quality Assurance, and Engineering Principles
 - System Configuration and Administration
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SKILLS

- Leadership and team collaboration across software, innovation, and community initiatives
 - Technical communication for documentation, project coordination, and stakeholder engagement
 - Problem solving and critical thinking in software engineering, networking, and system design
 - Strategic thinking and initiative in architecture planning, project execution, and innovation programs
 - Systems thinking and analytical decision-making
 - Adaptability, continuous learning, and emerging technology adoption
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PROJECTS & ENGINEERING EXPERIENCE

BIT System — Backend Developer & System Architect

Technologies: Python (Flask), MySQL, SQLite, JWT, REST APIs, HTML, CSS

Timeline: 2024 – Present

- Designed and developed a campus digital platform supporting student services, event management, payments, analytics, and administrative workflows.
- Implemented JWT-based authentication and role-based access control (RBAC) to secure system access and user permissions.
- Built RESTful APIs supporting structured data exchange and platform operations.
- Designed normalized relational database architectures to improve scalability, maintainability, and query efficiency.
- Integrated backend services with administrative tools and operational dashboards.

LearnLight — Software Developer

Technologies: React, JavaScript, APIs, Backend Services

Timeline: 2026

- Contributing to the development of a digital learning platform focused on structured online education and accessible learning experiences.
- Developing frontend and backend components supporting content delivery, user engagement, and platform functionality.
- Implementing API integrations and modular architecture to improve scalability and maintainability.
- Applying clean architecture principles and modern development practices throughout the platform.

Campus Pulse — Software Developer

Technologies: TypeScript, PL/SQL, JavaScript, CSS

- Developed a university engagement platform connecting students with campus events, hackathons, sports tournaments, career fairs, worship events, and student activities.
- Built features supporting event discovery, participation, registrations, and community engagement.
- Designed systems aimed at improving communication between event organizers and students.

SmartWaste — Software Developer

Technologies: Python, Flask, MySQL, REST APIs

- Developed a smart waste management platform designed to improve waste collection visibility, reporting, and operational efficiency.
- Built backend APIs for waste collection tracking, analytics, and data management.
- Designed relational database structures supporting operational reporting and decision-making.
- Applied data-driven approaches to improve monitoring of waste management activities and service delivery.

SkillBridge Africa (Africa Rise Ecosystem) — Software Developer

Technologies: TypeScript, PL/SQL

- Contributed to the development of a digital ecosystem connecting learning, mentorship, skills development, innovation, and real-world opportunities for African students and young innovators.
- Built platform features supporting opportunity discovery, structured learning pathways, mentorship access, and ecosystem connectivity.
- Helped bridge the gap between education, skills acquisition, and career opportunities through technology-driven solutions.

Nairobi Hidden Gems — Software Developer

Technologies: HTML, CSS, JavaScript

- Developed a social discovery platform helping users explore aesthetic locations, experiences, and hidden attractions across Nairobi.
- Focused on location-based engagement, user discovery experiences, and community-driven content exploration.

My Sim League — Software Developer

Technologies: TypeScript, JavaScript, CSS

- Built a football management simulation platform providing a realistic managerial experience across Europe's top football leagues.
- Designed gameplay systems, progression mechanics, league structures, and user engagement features.
- Focused on simulation logic, game architecture, and long-term player retention mechanics.

CERTIFICATIONS

Introduction to Modern AI — Cisco Networking Academy (2026)

- AI fundamentals
- Machine learning concepts
- AI applications and use cases

Junior Cybersecurity Analyst — Cisco Networking Academy

- Threat analysis
- Vulnerability assessment
- Security controls
- Security Operations Center (SOC) fundamentals

Huawei Computer Networks (CRA Training Program)

- Network architecture
- TCP/IP
- Routing and switching
- Network troubleshooting

Cisco Certified Cybersecurity Essentials

- Cyber risk management
- Secure access control

- Incident response

Data Science and Analytics — HP LIFE (2025)

- Data analysis
- Data visualization
- Reporting and analytical decision-making

AI Fluency: Framework and Foundations — Anthropic (2025)

- Prompt engineering
- AI workflows
- AI-assisted software development

Anticipatory Governance: Systems Thinking and Strategic Foresight for SDG-Aligned Integrated Planning – Part 1 (2026)

- Foundations of anticipatory governance
- Systems thinking frameworks

Anticipatory Governance: Systems Thinking and Strategic Foresight for SDG-Aligned Integrated Planning – Part 2 (2026)

- Strategic foresight implementation
- Applied systems thinking

SEO — HubSpot Academy (2025)

- Search engine optimization
- Search analytics
- Performance measurement

Content Marketing — HubSpot Academy (2024)

- Content strategy
- Audience engagement
- Digital storytelling

Digital Marketing — HubSpot Academy (2024)

- Campaign strategy
 - Social media analytics
 - Conversion optimization
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VOLUNTEER LEADERSHIP & PROFESSIONAL DEVELOPMENT

Aspire Leaders Program — Cohort 1 Graduate (2026)

- Completed a global leadership development program focused on leadership, systems thinking, problem solving, and professional growth.

Aspire Leaders Program — Moderator, Cohort 2 (2026)

- Facilitate participant engagement, discussions, and peer support activities across learning cohorts.
- Support program operations and community development initiatives.

GMI Mentorship Program (2026)

- Participated in structured mentorship focused on leadership development, career growth, and professional excellence.

Secretary — Innovation Club, Zetech University

- Support innovation initiatives, student engagement programs, and technology-focused activities.
- Coordinate club operations, communication, and project activities.